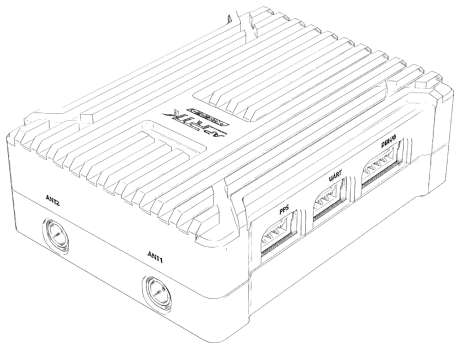


AP-RTK X20D

User Guide



AP-RTK X20D Overview



One u-blox ZED-X20D all-band receiver and two antennas provide RTK position and heading, forwarded with RM3100 compass data to the flight controller over DroneCAN (engineering evaluation).

AP-RTK X20D Specifications

• Basic Specifications

Receiver	u-blox ZED-X20D — single module, dual antenna, HDG 2.00
GNSS Bands	All-band GNSS on both antennas (L1 / L2 / L5 / L6)
MCU	STM32F412Rx, ARM Cortex-M4, 512 KB flash
Compass	PNI RM3100 on I2C3 — axis qualification pending
Heading	Receiver-computed ANT1-to-ANT2 heading; UBX-NAV-DAHEADING v2
Update Rate	Firmware requests 5 Hz navigation at 230400 baud; end-to-end CAN rate requires measurement

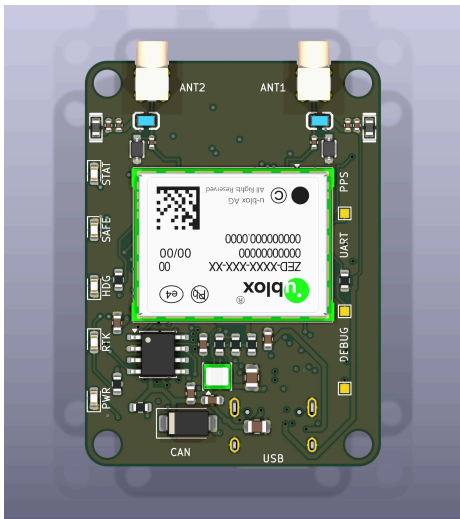
AP-RTK X20D Specifications

• Basic Specifications

Comm. Protocol	DroneCAN — position, ardupilot.gnss.Heading and separate magnetometer data
I/O Ports	2× MMCX antenna · CAN · USB-C (MCU) · external receiver UART
PCB	R3 · 33.00 × 45.17 mm · 4 layers · 0.8 mm
Firmware	AP_Periph · novaX 1.0.1 engineering release · board ID 6205
Validation Status	Windows GCC 10.2.1 build verified; first-article hardware and flight qualification pending

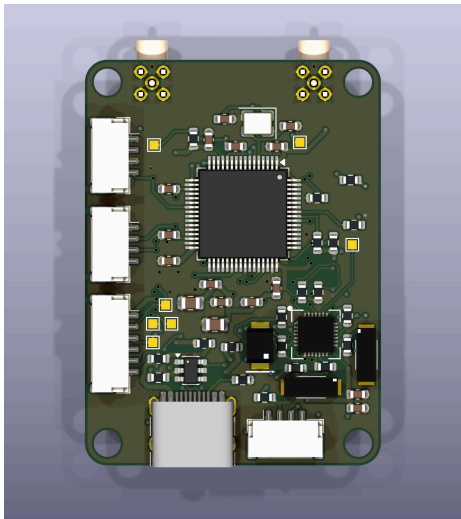
AP-RTK X20D Pinout

- Pinout



AP-RTK X20D Pinout

- Pinout



AP-RTK X20D Pinout

• Pinout Notes

R3 PCB connector-location renders, top and bottom. USB-C connects to STM32F412 OTG_FS for MCU service / bootloader / DFU; it is not the ZED-X20D USB port. The receiver communicates with the MCU through USART2 (GPS port index 4). PC6/PC7 external UART MCU TX/monitor is not implemented in v1.0.1. Use the R3 schematic to verify connector pin numbering and supply before first power-on; do not infer the pinout from AP-RTK dual or G5H.

AP-RTK X20D Setup

1. **Antenna direction** — ANT1 (RF_IN_1) at the rear, ANT2 (RF_IN_2) at the front, on the centerline at equal height; measure the baseline.
2. **H7E connection** — Connect over DroneCAN and set the parameters below. No X20D-specific H7E firmware build is needed.
3. **Position offset** — Measure ANT1 relative to the centre of gravity and enter GPS1_POS_X/Y/Z (+X forward, +Y right, +Z down).
4. **First-article checks** — On the bench, confirm receiver startup, RTK corrections, position, heading at known directions, reacquisition, RM3100 axes, power and update paths.

AP-RTK X20D Setup

• Flight Controller Parameters

Parameter	Value	Description
CAN_P1_DRIVER	1	Enable CAN1; reboot after changes
CAN_D1_PROTOCOL	1	DroneCAN
GPS1_TYPE	9	DroneCAN GPS (GPS_TYPE on older firmware)
GPS1_MB_TYPE	0	No moving-baseline offset conversion; receiver already supplies forward heading
AHRS_EKF_TYPE	3	Use EKF3
EK3_ENABLE	1	Enable EKF3
EK3_SRC1_YAW	2	GPS yaw; use 3 for compass fallback only after the RM3100 axes and calibration have been qualified

AP-RTK X20D Firmware

• Available Firmware

Type	Version	Date
AP_Periph (DroneCAN OTA)	1.0.1	2026-09-07
AP_Periph (.apj package)	1.0.1	2026-09-07
Bootloader + App (merged HEX)	1.0.1	2026-09-07
Release verification (JSON)	1.0.1	2026-09-07

Download files from the product page (QR).

AP-RTK X20D Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.
- Firmware 1.0.1 is an engineering release — do not fly before first-article hardware and flight qualification.
- Antenna direction is opposite to AP-RTK dual (ANT1 rear, ANT2 front); swapped cables give a 180° heading error.
- Board ID 6205 only — never flash dual (1085) or G5H (6206) images.

AP-RTK X20D Safety Notes



<Detailed Manual>

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